star gets too close to this monster, is disintegrated by the gravitational distortion and swallowed by the cyclopean black hole, like a small fish swallowed by a whale. Imagine a monster the size of a million Suns, which swallows in an instant our own Sun and its miniature planets …

There is a wonderful ongoing project to construct a network of radio antennae distributed across the Earth from pole to pole, with which astronomers will be able to achieve a resolution sufficient to 'see' the galactic black hole. What we expect to see is a small black disc surrounded by the light produced by the radiation of the matter falling in.

What enters a black hole does not come out again, at least if we neglect quantum theory. The surface of a black hole is like the present: it can be crossed only in one direction. From the future, there is no return. For a black hole, the past is the outside; the future is the inside. Seen from outside, a black hole is like a sphere which can be entered but out of which nothing can come. A rocket could stay positioned at a fixed distance from this sphere, which is called the *horizon* of the black hole. To do so it needs to keep its engines firing intensely, to resist the gravitational pull of the hole. The powerful gravity of the hole implies that time slows down for this rocket. If the rocket stays near enough to the horizon for one hour, and then moves away, it would then find that, outside, in the meantime, centuries have passed. The closer the rocket stays to the horizon, the slower – with respect to the outside – time runs for it. Thus, travelling to the past is difficult, but travelling to the future is easy: we need only to get close to a black hole with a spaceship, keep within its vicinity for a while, and then move away.

On the horizon itself, time stops: if we get extremely close to it and then move away after a few of our minutes, a million years might have elapsed in the rest of the universe.

The really surprising thing is that the properties of these strange objects, today commonly observed, were *foreseen* by Einstein's theory. Now, astronomers study these objects in space, but until not long ago black holes were considered a scarcely credible and bizarre consequence of an outlandish theory. I remember my university professor introducing them as solutions to Einstein's equations, to which 'real objects were unlikely ever to correspond'. This is the stupendous capacity of theoretical physics to discover things before they are observed.

The black holes we observe are well described by Einstein's theory, and quantum mechanics is not needed to understand them. But there are two mysteries of black holes that do require quantum mechanics in order to be unravelled and, for each of these, loop theory offers a possible solution. One of these could also offer an opportunity to test the theory.

The first application of quantum gravity to black holes concerns a curious fact discovered by Stephen Hawking. Early in the 1970s he theoretically